

HRISHIKESH BELAGALI

Undergraduate Student

📍 East Lansing

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SUMMARY

Undergraduate computer science major with research interests in quantum algorithms, molecular dynamics, reservoir computing, quantum dynamics, and high performance computing.

SKILLS

Languages: Python, C, C++, $\text{\LaTeX}$

Packages: NumPy, PyTorch, scipy, Qiskit, Cirq, PyEMMA, hyperopt, VMD-Python

Softwares: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S)

Expected Graduation: 2027

Current GPA: 4.0

Michigan State University Honors College

EXPERIENCE

2025–

Honors Research Scholar

MSU Honors College/MSU-Q

- Conducted research on generalized quantum simulators under the guidance of Professor Ryan LaRose.
- Investigated computational group theory methods for subgroup membership in unitary subgroups.
- Implemented compilation optimization for generalized subgroups using Python and Cirq.

2024–

Undergraduate Research Assistant

Vermaas Lab, Plant Research Laboratory

- Investigated the interaction of the chloroplast targeting protein (At4g16060) with thylakoid membranes using molecular dynamics.
- Analyzed the effect of PDLP-5 (Plasmodesmata Located Protein) on water flow rates through PIP1;4 aquaporin tetramer using grid-steered molecular dynamics.
- Simulated the effects of small molecules like butanol on *Clostridium acetobutylicum* and *Zymomonas mobilis* membrane dynamics.

COURSEWORK

- Advanced Linear Algebra
- Data Structures and Algorithms, Programming in C++, Computer Architecture
- Mechanics, Electricity and Magnetism, Thermodynamics and Modern Physics, Classical Mechanics
- Graduate Quantum Algorithms (Scheduled)
- Graduate Computational Foundations of Artificial Intelligence (Scheduled)

CONFERENCES

2025

PRL, MSU

Aquaporin Water Transit Modulation via Protein Capping

acs.digitellinc.com

Abstract presented at ACS Fall 2025 Conference (contributed)

2025

PRL, MSU

Evaluating the Interaction of Chloroplast-targeting Protein (At4g16060) with Thylakoid Membrane

Abstract presented at First Great Lakes Plant Science Conference (contributed)

PROJECTS

Python

Qiskit

Discretized Quantum Adiabatic Evolution

github.com

Implemented a quantum adiabatic algorithm to solve QUBO type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques

Python

Qiskit

Quantum Arithmetic Circuits

github.com

Built quantum circuits to perform addition, subtraction and multiplication using the Quantum Fourier Transform.

Python

NumPy

Monte Carlo Integration

github.com

Monte Carlo integration with importance sampling to numerically evaluate complex integrals.

Python

NumPy

Langevin Monte Carlo

github.com

Metropolis-adjusted Langevin algorithm for sampling from intractable probability distributions.

Python OVITO pyMOL	Genetic Annealing to Determine Protein Structures github.com Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation ($\text{RMSD} < 3.0$).
Python	Quadratic Unconstrained Binary Optimization github.com Stochastic tunneling-enhanced simulated annealing for solving QUBO-type problems.
Python NumPy	aqua-blue github.com/pypi.org Secondary maintainer of aqua-blue, a package building echo state networks. • Implemented reservoir state time-lag, sparsity and spectral radius to improve prediction accuracy. • Introduced timezone-aware datetime support for seamless interfacing with data pipelines. • Demonstrated prediction capability by predicting quantum time evolution of Gaussian wave packets.
Python NumPy hyperopt	aqua-blue-hyperopt github.com/pypi.org Primary maintainer of aqua-blue-hyperopt, a package implementing hyperparameter optimization functionality for aqua-blue models. • Created modular hyperparameter optimization functionality for aqua-blue reservoir computers. • Implemented custom loss function capabilities for flexible model evaluation.

OPEN SOURCE CONTRIBUTIONS

QuTiP Python	Quantum Toolbox in Python (QuTiP) github.com Contributed to QuTiP, an open-source software for simulating quantum systems. • Implemented functionality to return eigenkets as an operator in the Qobj class.
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HONORS AND AWARDS

- Wielenga Research Scholarship, Honors College (2025-2026)
- Dean's List (Fall 2024, Spring 2025, Fall 2025)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024)
- MSU President's Scholarship (2024)